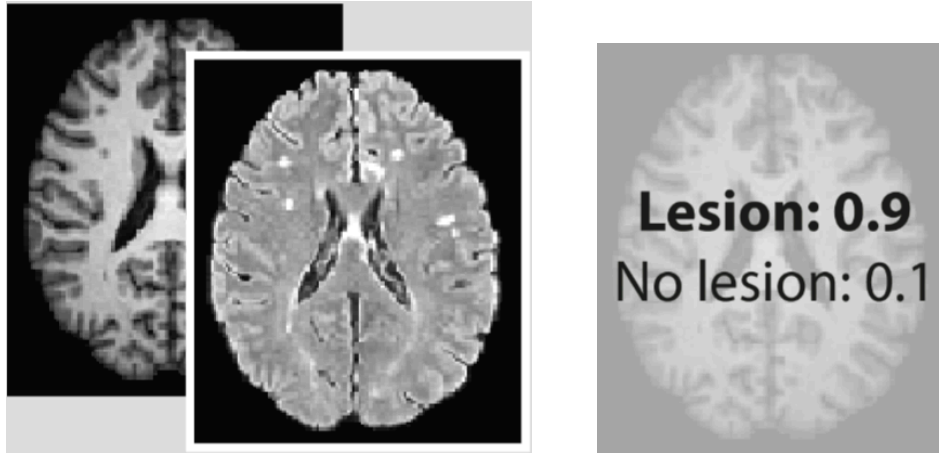


Use Case 3: Identification of new multiple sclerosis lesions in follow-up MRI scans



You are part of a research team developing an AI model to identify new multiple sclerosis (MS) lesions in longitudinal multi-modal brain MRI scans. The goal is to support neurologists in monitoring disease progression and assessing treatment response over time.

The system analyzes follow-up MRI examinations and predicts whether newly appearing lesions are present. For each case, the model outputs a confidence score for every possible class, and the final prediction is obtained by selecting the class with the highest predicted score. During deployment, clinical decisions are triggered once prediction scores exceed predefined operating thresholds.

In clinical practice, missing newly emerging lesions may lead to delayed recognition of disease progression and inappropriate treatment decisions. At the same time, falsely identifying lesion activity may result in unnecessary follow-up procedures, patient anxiety, or treatment escalation. Clinicians therefore consider some types of classification errors substantially more critical than others, and the estimated confidence of the model predictions is important for decision-making.

The dataset reflects the distribution of cases typically encountered in the target population (prevalence). However, truly new lesions occur considerably less frequently than stable findings, meaning that positive cases are relatively rare. For probability assessment, the team prefers metrics with straightforward interpretation and bounded behavior, as the results will later be communicated to clinicians and clinical researchers.

The clinical team is particularly interested in balancing the reliable identification of new lesion activity with the avoidance of excessive false positive findings. Since lesion activity is relatively rare, clinicians are especially interested in the reliability of positive predictions.

You now want to validate the model appropriately and decide which metrics should be used for reporting model performance.

Task description

Study your use case carefully. Then go to: <https://metrics-reloaded.dkfz.de/metric-selection>

Follow the decision process of the Metrics Reloaded toolkit and answer the fingerprint questions based on the information provided in the given scenario. Your goal is to identify the most appropriate performance metrics for the given application.

After finishing the task, save the generated PDF report containing the selected metrics and fingerprint answers. Be prepared to briefly present your use case to the group, including:

- how you interpreted the scenario,
- which fingerprint questions were easy or difficult to answer,
- whether any design choices were unclear or ambiguous,
- and which metrics were ultimately selected by the toolkit.

Please send your PDF report to the lecturers after completing the exercise.